

PRODUCT DESIGN & OPERATING SPECIFICATION

New Connection Management

A page-based, auditable workflow for application intake, inspection, quotation, payment, approval, customer conversion, installation and service activation.

Samdamte Water Utility Management

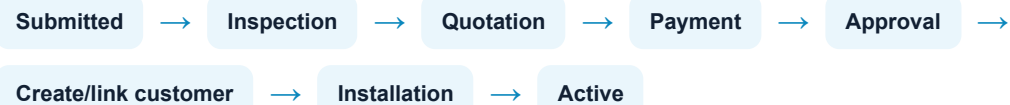
Version 1.0 · 27 July 2026

One application, one visible lifecycle

Existing design system · responsive · server-paginated

Operating flow

Each status unlocks only the next valid action. All decisions and changes create an activity record.



Customer integration: the existing Create Customer wizard is opened with the connection application context and returns to the application after linking the new customer. It is reused, not duplicated.

Interaction rules

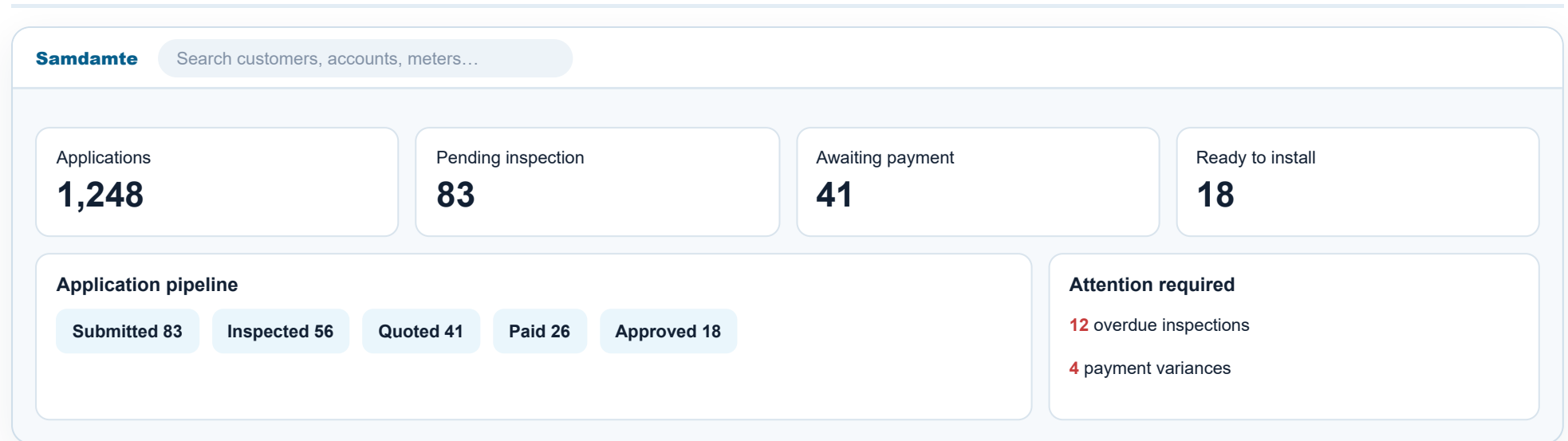
Use full pages for the multi-step intake and application profile. Use right drawers or confirmation dialogs for short actions, fee override, destructive decisions and error detail.

Lists load 25 records by default and filter on the server.

SCREEN 1

Connection dashboard

Workload, conversion and ageing at a glance.

[+ New application](#)

SCREEN 2

New connection application

A focused four-step page wizard.

Save draft

New Connections New application

1 Applicant

2 Property

3 Service & fee

4 Review

Applicant type

New applicant

Applicant name *

Enter full legal name

Phone *

+254...

Email

name@example.com

Application guidance

Capture the applicant first. A customer account is created only after approval to prevent incomplete records entering the customer register.

Continue

Application register

Searchable, server-paginated operational register.

+ New application

Application no., applicant or phone

All statuses

All zones

APPLICATION	APPLICANT	LOCATION	FEE	STATUS	ACTION
NC-2026-000124	Grace Njeri +254 712 000 000	Joska 3 · Route 01	KSh 2,000.00	Submitted	Review →
NC-2026-000123	Peter Maina	Baraka · Route 02	KSh 2,000.00	Inspection scheduled	Review →

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Application profile

Single source of truth for the connection journey.

Inspection scheduled

Application

Inspection

Quotation

Payment

Approval

Installation

Activation

NC-2026-000124 · Grace Njeri

Plot 18, Joska 3 · Domestic connection · +254 712 000 000

Activity

Application submitted	27 Jul 2026 09:41	Customer Care
Inspection scheduled	28 Jul 2026 10:00	Mary Supervisor

Next action

Record the field inspection outcome and technical notes.

Record inspection

SCREEN 5

Inspection scheduling

Short contextual action on the application profile.

Schedule field inspection

Date and time *

28/07/2026 · 10:00 AM

Inspection officer

Search field officers...

Instructions

Verify service line, access and required materials

Schedule inspection

Confirmation drawer

Use a right drawer only to confirm the officer, date and applicant notification before committing.

SCREEN 6

Inspection report

Technical assessment with a clear outcome.

Outcome *

Feasible

Technical findings *

Existing main is accessible. 12 metres of 20 mm pipe required.

Evidence

Upload photos or documents

Outcome rules

Feasible unlocks quotation.

Revisit returns to scheduling.

Not feasible requires a reason and closes the application.

Save report

Quotation

Transparent calculation and controlled fee override.

ITEM	BASIS	AMOUNT
Connection fee	System default	KSh 2,000.00
Materials	Inspection estimate	KSh 4,850.00
Labour	Standard job rate	KSh 1,500.00
Total		KSh 8,350.00

Fee policy

The default connection fee is configured once in Settings and copied to the application at submission.

Override is available to authorized users and requires a reason. The original and overridden amounts remain auditable.

Override fee

Issue quotation

Fee and payment tracking

Receipts, partial payments and outstanding balance.

Quotation

KSh 8,350

Paid

KSh 5,000

Balance

KSh 3,350

Status

Partially paid

Payment history

DATE	REFERENCE	CHANNEL	AMOUNT
29 Jul 2026	QHL8D2K3	M-Pesa	KSh 5,000.00

Record payment

Reference

Amount

Record payment

SCREEN 9

Approval or rejection

Independent decision after full payment.

Approval pack

Applicant details	Complete
Inspection	Feasible
Quotation	KSh 8,350.00
Payment	Paid in full

Decision comments *

Record the reason for this decision

Reject

Approve application

SCREEN 10

Create or link customer account

Reuse the existing Create Customer workflow.

Approved applicant is ready for conversion

Choose an existing customer if this applicant already has a profile, or launch the current Create Customer wizard with application data prefilled.

Link existing customer

Open Create Customer

The existing wizard receives **connectionId** and **returnTo**. After its normal create operation, it links the resulting customer and returns to this profile.

Do not duplicate

Name, contact, identity and customer validation remain owned by the established customer module.

SCREEN 11

Installation work order

Handoff to the existing Work Orders module.

Connection installation

Customer account

Zone / route

Assign field officer

Dispatch summary

New connection · High priority

Due 02 Aug 2026

Create work order

SCREEN 12

Installation completion

Confirm physical work before activation.

Completion evidence

Meter serial *

SDM-0001842

Initial reading *

0

Completion notes *

Connection tested; no leaks observed.

Verification

Installation is marked complete first. Activation remains a separate controlled action.

Complete installation

Customer notifications

Event-driven updates across the journey.

EVENT	RECIPIENT	CHANNEL	DELIVERY
Application received	Grace Njeri	SMS	Delivered
Quotation issued	Grace Njeri	Email + SMS	Delivered
Connection activated	Grace Njeri	SMS	Queued

Notification failures do not roll back operational work; they enter the existing delivery queue for retry and audit.

Status tracking and activation

A final, readable view for staff and customer follow-up.

Active

Application

Inspection

Quotation

Payment

Approval

Installation

Activation

Connection successfully activated

Customer: Grace Njeri · 10017

Meter: SDM-0001842 · **Activated:** 03 Aug 2026 14:32

Application: NC-2026-000124

Available actions

Open customer profile

Open meter profile

View work order

Print application record

IMPLEMENTATION NOTE

Fee ownership and deployment

Connection fee decision

Implemented assumption: the fee is configured once under Settings → Billing & connections, then copied into each application at submission. A per-application override is supported and requires a reason.

If no default has been configured, the interface displays **Not configured** and prevents an accidental zero-fee application. An authorized user may still enter an explicit override and reason.

This does not conflict with an existing connection implementation because the repository did not previously contain a complete new-connection fee workflow. It does coexist with tariff and billing charges: the connection fee is an application/installation charge, not a consumption tariff.

Live deployment

Deploy code, then run the committed database upgrade and regenerate the client. Do not manually recreate the tables.

```
npm run db:connection-upgrade  
npm run prisma:generate  
npm run build
```